

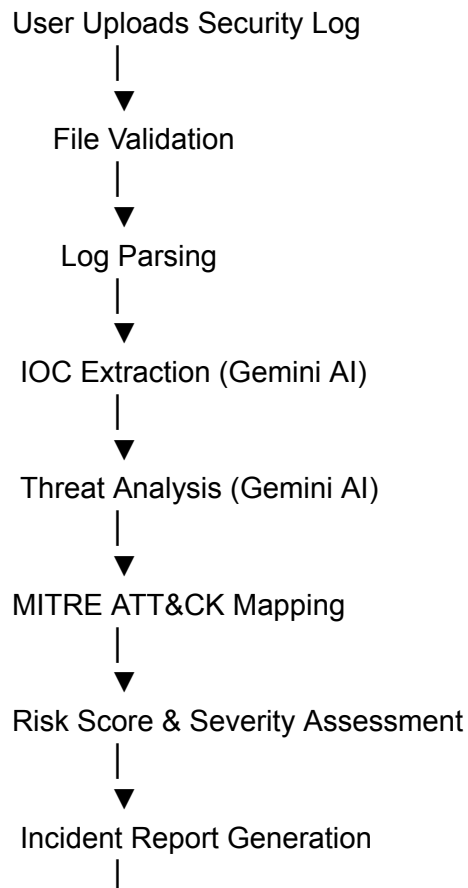
Incident Response Workflow

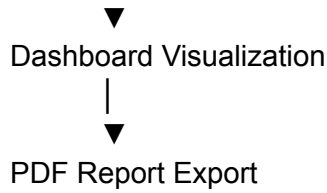
Objective

The Incident Response Workflow defines the sequence of operations performed by the AI SOC Copilot after a security log is uploaded. The workflow combines log parsing, AI-powered IOC extraction, threat analysis, MITRE ATT&CK mapping, severity assessment, and automated report generation to assist Security Operations Center (SOC) analysts in responding to cybersecurity incidents efficiently.

Workflow Overview

The AI SOC Copilot follows an automated incident response pipeline to analyze uploaded security logs and generate actionable threat intelligence.





Workflow Description

Step 1 – Log Upload

The user uploads a supported security log file through the web interface. The application currently accepts TXT, JSON, and CSV log formats.

Step 2 – File Validation

The uploaded file is validated before processing. Invalid or unsupported files are rejected to ensure only valid security logs are analyzed.

Step 3 – Log Parsing

The parser converts raw security logs into structured events that can be understood by the AI analysis modules.

Typical information extracted includes:

- Event timestamps
 - Process names
 - User activity
 - IP addresses
 - Network events
 - Authentication events
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Step 4 – IOC Extraction

The parsed logs are provided to the Gemini Large Language Model to automatically extract Indicators of Compromise (IOCs).

Supported IOC categories include:

- IPv4 addresses
- IPv6 addresses
- Domains
- URLs
- Email addresses
- File hashes
- CVE identifiers

The extracted IOCs are returned as structured JSON objects.

Step 5 – Threat Analysis

The parsed logs and extracted IOCs are analyzed by Gemini AI to determine:

- Attack Type
 - Severity Level
 - Risk Score
 - Confidence Score
 - MITRE ATT&CK Mapping
 - Technical Reasoning
 - Recommended Actions
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Step 6 – MITRE ATT&CK Mapping

Based on the identified attack, Gemini maps the incident to the corresponding MITRE ATT&CK tactic, technique, and technique ID. This provides analysts with a standardized understanding of attacker behavior.

Step 7 – Severity Assessment

The AI assigns a Risk Score (0–100) and Severity Level (Low, Medium, High, or Critical), allowing analysts to prioritize incidents based on potential impact.

Step 8 – Incident Report Generation

Using the threat analysis results and parsed logs, the AI generates a comprehensive incident report containing:

- Executive Summary
 - Technical Summary
 - IOC Summary
 - MITRE ATT&CK Mapping
 - Timeline of Events
 - Containment Recommendations
 - Recovery Steps
 - Analyst Notes
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Step 9 – Dashboard Visualization

The generated results are displayed in the dashboard, allowing analysts to review:

- Threat Summary
 - Severity
 - Risk Score
 - MITRE Mapping
 - Timeline
 - IOC Summary
 - Recommended Actions
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Step 10 – PDF Export

The completed incident report can be exported as a PDF for documentation, incident tracking, or sharing with stakeholders.

Benefits of the Workflow

The automated workflow provides several advantages:

- Faster security log analysis.
- Automated IOC extraction.
- AI-assisted threat classification.
- Standardized MITRE ATT&CK mapping.
- Consistent incident reporting.

- Improved SOC analyst productivity.
 - Reduced incident response time.
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Conclusion

The AI SOC Copilot automates the complete incident response lifecycle, from log ingestion to report generation. By integrating AI-driven IOC extraction, threat analysis, MITRE ATT&CK mapping, and incident reporting, the workflow supports efficient and consistent security operations.